RAG Hallucination Evaluation in Financial Domain

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1. Background

Retrieval-Augmented Generation (RAG) systems enhance Large Language Models (LLMs) by grounding responses in external knowledge bases, reducing reliance on parametric memory.

However, RAG systems remain susceptible to hallucinations — generating plausible but factually incorrect or unsupported information. This is especially critical in high-stakes domains such as finance.

This project builds and evaluates three RAG architectures over a cleaned Reuters finance corpus, with a focus on detecting and mitigating hallucinations through self-reflection mechanisms.

2. Motivations & Objectives

Motivations

- Standard RAG systems lack self-correction, leading to unverified outputs.
- Financial queries demand high factual accuracy and strict source grounding.
- No unified benchmark compares dense, lexical, and self-reflective RAG.

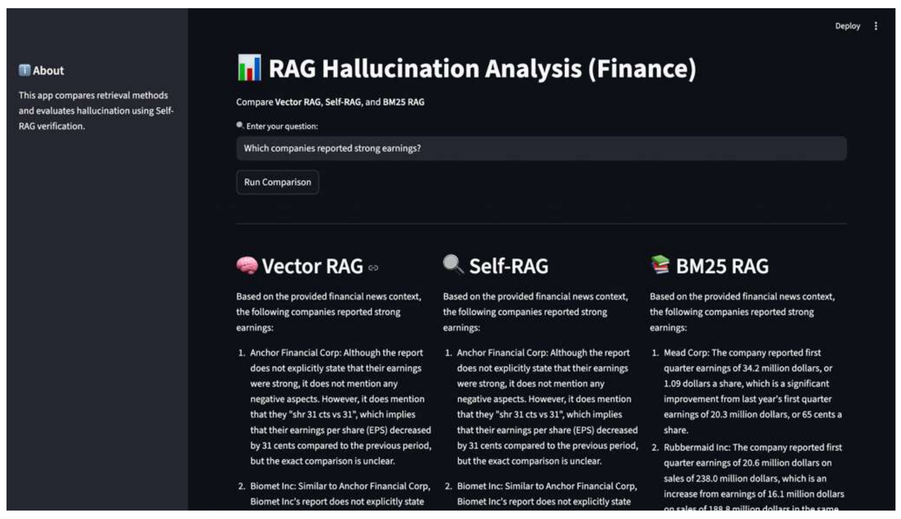
Objectives

- Build and compare Vector RAG, BM25 RAG, and Self-RAG on a finance corpus.
- Implement Self-RAG with LLM-as-a-Judge grading for relevance, groundedness, and usefulness.
- Develop a Streamlit UI for side-by-side comparison of all three approaches.
- Quantify hallucination rates and self-correction effectiveness via batch evaluation.

3. Related Work

RAGAS: Reference-free evaluation framework for grounding and relevance. https://arxiv.org/abs/2309.15217
Self-RAG: Self-reflective retrieval and generation using critique tokens. https://arxiv.org/abs/2310.11511
CRAG: Benchmark for evaluating RAG robustness against noisy retrieval. https://arxiv.org/abs/2406.04744
Cleanlab: Benchmarking study on hallucination detection in RAG systems. https://cleanlab.ai/blog/rag-tlm-hallucination-benchmarking/
HaluEval: Large-scale dataset for evaluating hallucinations in LLMs. https://arxiv.org/abs/2305.11747

4. Methodology



Dataset

Cleaned Reuters finance corpus (finance_reuters_cleaned.csv) containing financial news articles. Text is chunked and indexed for both dense and lexical retrieval.

Architectures Implemented

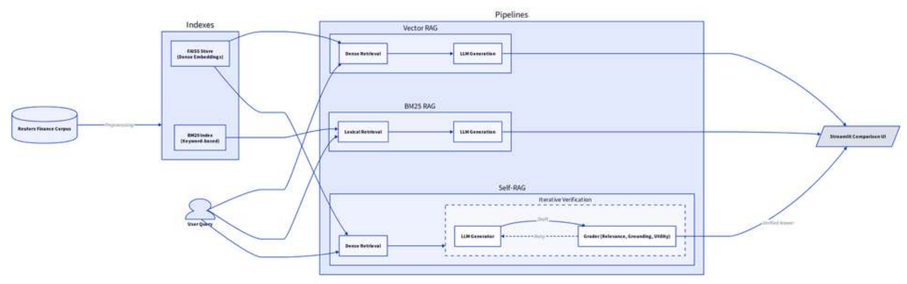
1. Vector RAG (Baseline): FAISS dense retrieval (all-MiniLM-L6-v2) + Groq LLM (llama-3.1-8b-instant) for generation.
2. BM25 RAG: BM25 lexical index retrieval + Groq LLM for generation, with BM25 relevance scores displayed.
3. Self-RAG: Extends Vector RAG with three LLM-as-a-Judge verification steps:
   a) Retrieval Grader: Is the retrieved context relevant to the query? (YES/NO)
   b) Hallucination Grader: Is the answer grounded in the context? (YES/NO)
   c) Answer Grader: Does the answer address the user's question? (YES/NO)
   If any check fails, the system retries retrieval and generation automatically.

Evaluation Pipeline

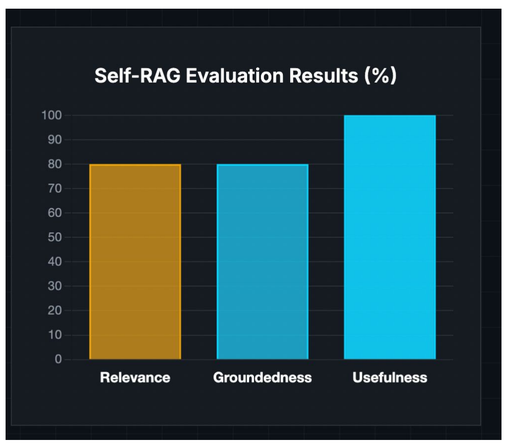
Batch scripts run 10 predefined financial queries across all models. Self-RAG logs Relevance, Groundedness, Usefulness, and Retry counts per query to artifacts/eval/.

Technology Stack

Python | FAISS | BM25 | sentence-transformers | Groq API (llama-3.1-8b-instant) | Streamlit | dotenv



5. Experimental Results



Self-RAG Evaluation on 10 Financial Queries

Metric	Result
Context Relevance	80.0% (8 / 10 queries)
Answer Groundedness	80.0% (8 / 10 queries)
Answer Usefulness	100.0% (10 / 10 queries)
Average Retries	0.70 per query

Self-Correction Analysis

- 60% of queries (6/10) passed all checks on the first attempt with zero retries.
- 40% of queries (4/10) triggered the self-correction loop (1-2 retries), indicating the grader successfully detected initial hallucinations or irrelevant context.
- All 10 queries ultimately produced useful answers, demonstrating the robustness of the pipeline.

Qualitative Observations

- Vector RAG and BM25 RAG produced answers without verification, occasionally including ungrounded claims.
- Self-RAG correctly identified out-of-domain queries (e.g., cryptocurrency regulations) and flagged them as not grounded, avoiding fabricated responses.
- BM25 RAG excelled on keyword-heavy financial queries; Vector RAG performed better on semantic queries.

Conclusion

Self-RAG reduces hallucination risk by verifying both retrieval quality and answer grounding, achieving 100% usefulness while maintaining 80% groundedness across diverse financial queries.

Future work: Evaluate performance on larger LLMs, Integrate Graph RAG for structured retrieval comparison, Relax baseline RAG constraints for stronger control benchmarking, Increase evaluation diversity to capture broader system behavior